



Sumitomo Heavy Industries, Ltd.

MANUAL NUMBER: CD32ZZ-098D

DATE: March 17 / 2003

TECHNICAL INSTRUCTION

RDK-408A2 4K COLD HEAD Revision 0

For Service Personnel Only

PART NUMBERS

MODEL	SHI PART No.	GE PART No.
RDK-408A2 4K COLD HEAD	RD42ZN1079	2218465-3

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CROSS REFERENCE

Thoroughly read this manual and following manuals before using this equipment.

MANUAL NAME	MANUAL No.
OPERATION MANUAL SRDK Series CRYOCOOLER	CD32ZZ-056
TECHNICAL INSTRUCTION CSA-71A COMPRESSOR UNIT*	CD32ZZ-060
TECHNICAL INSTRUCTION CSW-71C COMPRESSOR UNIT*	CD32ZZ-061
TECHNICAL INSTRUCTION CSW-71D COMPRESSOR UNIT*	CD32ZZ-062
INSTALLATION MANUAL BPU-01 BY PASS UNIT**	CD32ZZ-075

* See the TECHNICAL INSTRUCTION of Compressor Unit used.

** See the INSTALLATION MANUAL of BPU-01, if BPU-01 is used in this system.

1 GENERAL INFORMATION

RDK-408A2 Cold Head is a two-stage GM cycle cryo-refrigerator. The function of the Cold Head is to produce continuous closed-cycle refrigeration at temperatures, depending upon the heat load imposed, in the range of 25 K to 40 K for the first-stage cold station and in the range of 3.5 K to 4.2 K for the second-stage cold station.

The Cold Head has three major components: the drive unit; the cylinder; and the displacer-regenerator assembly, which is located inside the cylinder.

With newly developed rare earth regenerator material and with very unique structure, the model RDK-408A2 Cold Head has its 2nd stage refrigeration capacity of 1W at 4.2K.

Functionally, the high-pressure helium gas from the Compressor Unit will be supplied to the Cold Head through the helium gas supply connector. The supply gas will be passed into the displacer-regenerator assembly, come out through the displacer-regenerator assembly to the crankcase through the motor housing, and finally will be returned to the Compressor Unit through the helium gas return connector. The helium gas expansion in the displacer-regenerator assembly will be provided cooling condition for the first and second-stage cold stations.

1-1 SPECIFICATIONS

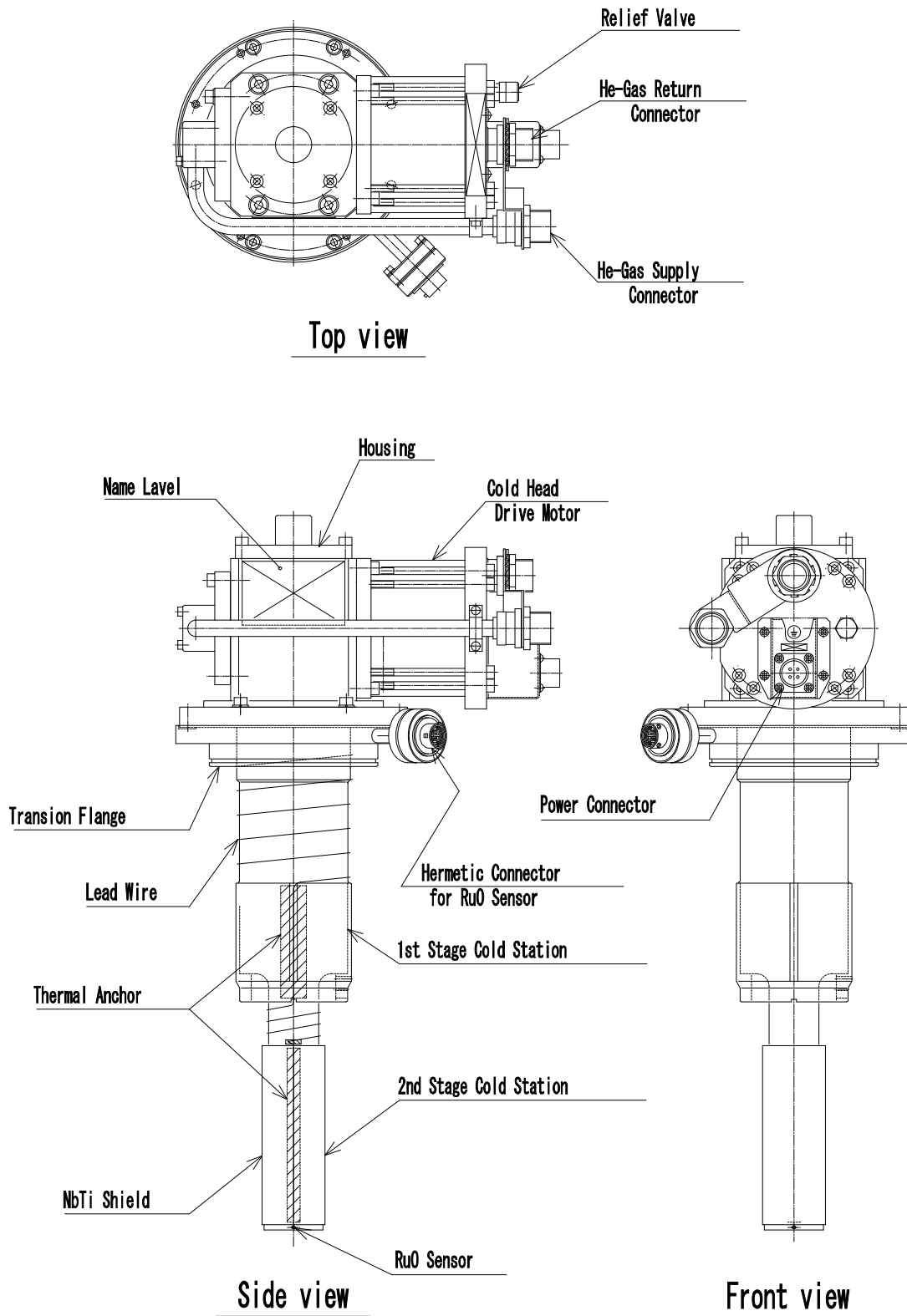
The specifications of RDK-408A2 Cold Head are summarized in **Table 1.1**

Figure 1.1 shows the outline view of Cold Head.

Table 1.1 RDK-408A2 COLD HEAD SPECIFICATION

Refrigeration Capacity First Stage Second Stage	44 W at 43 K (50 / 60 Hz) 1.0 W at 4.2 K (50 / 60 Hz)
Orientation	Any Capacity Loss: Max. 15 %
Ambient Operating Temperature	5 to 28 deg.C (41 to 82.4 deg.F) 28 to 35 deg.C (82.4 to 95 deg.F) with 5% Capacity Loss
Helium Gas Pressure Static Operating (High Side)*	1.60 - 1.65 MPa at 20 deg.C (68 deg.F) (16.3 - 16.8 kgf/cm ² G, 232 - 239 psig) 2.10 - 2.30 MPa --- approx. (21.4 - 23.5 kgf/cm ² G, 319 - 333 psig)
Pressure Relief Valve Setting	1.86 - 1.96 MPa (19.0 - 20.0 kgf/cm ² G, 270 - 284 psig)
Gas Supply Connector Gas Return Connector	1/2-inch Coupling 1/2-inch Coupling
Dimension Width Length Height	180 mm (7.09') 294 mm (11.58') 557 mm (21.93')
Weight	20.3 kg (44.7 LBS) --- approx.

* The operating pressure varies according to the heat load of cold head and temperature around the equipment.



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Figure 1.1 OUTLINE VIEW OF COLD HEAD MODEL RDK-408A2

1-2 CONSTRUCTION

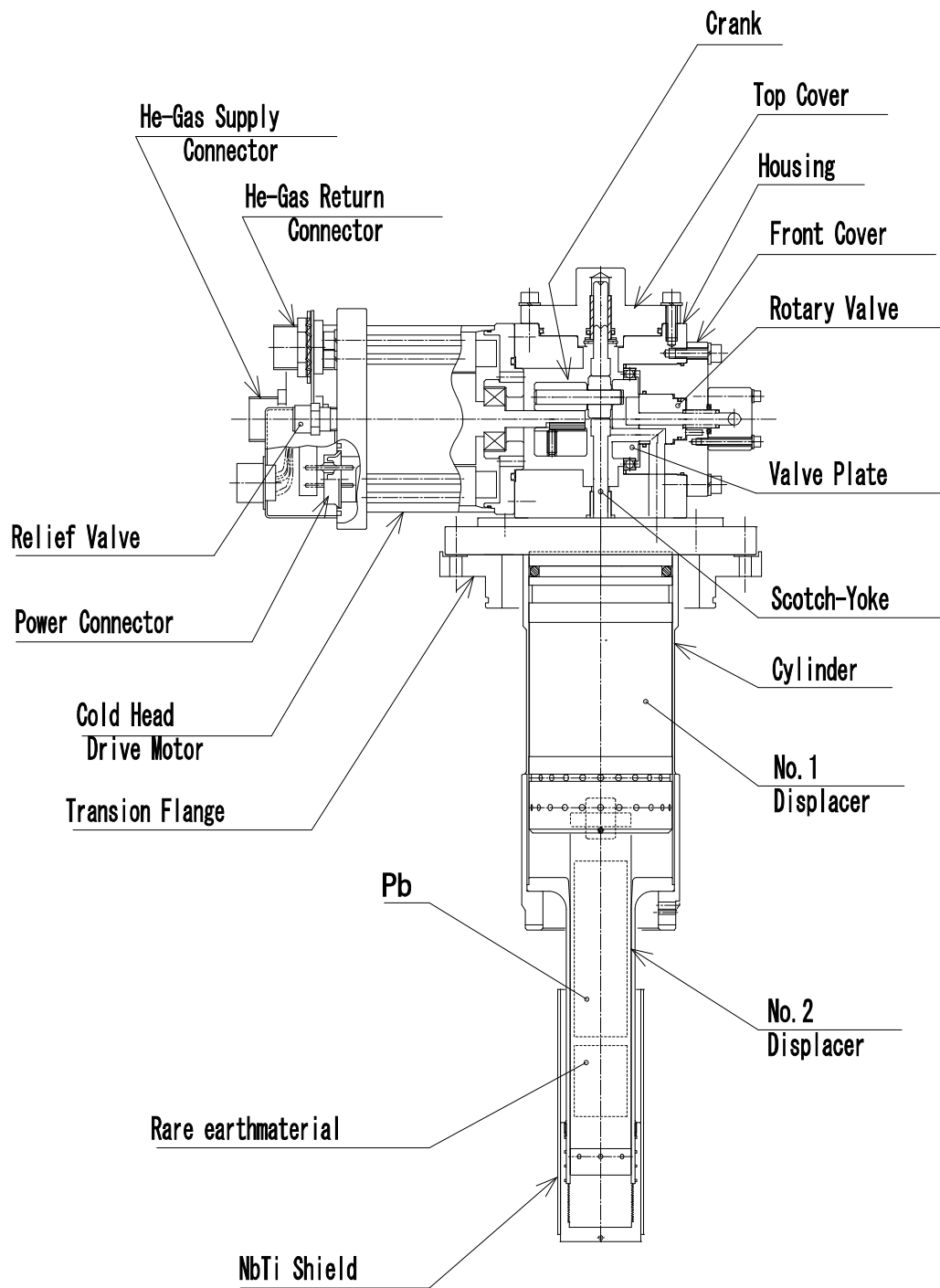
The cross section of RDK-408A2 Cold Head is shown in **Figure 1.2**. It consists of a Cylinder, No. 1 Displacer, No. 2 Displacer, drive mechanism, and Cold Head Drive Motor. No. 1 Displacer is connected to the Scotch Yoke which can be driven by the Cold Head Drive Motor through the Crank with Bush so that the rotation of the Cold Head Drive Motor can be varied to reciprocating motion of Scotch Yoke and Displacers.

The Rotary Valve system is furnished to control the helium gas intake and exhaust timing. The Rotary Valve is also coupled to the Cold Head Drive Motor through Crank, so intake and exhaust operation is synchronized with the position of the Displacer.

The Displacer is a loose fit in the Cylinder except at the top and where it is equipped with a dynamic(sliding) seal to prevent leakage passed through the clearance between the Displacer and Cylinder.

The Displacers consist of regenerator material which cool the gas when passing downwards to the cold space and heats the gas when passing upwards from the cold space. Rear earth regenerator material is used in the 2nd stage Displacer to produce the cooling capacity at the temperature of 4.2K.

The pressure above and below the Displacer is the same except for small pressure drops across the regenerator when gas is flowing through it. Virtually no physical work is required to move the Displacer in the Cylinder. No work is done on the gas and the gas does no work on the Displacer. The pressure in the system is increased or decreased by operation of the inlet or outlet valves.



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Figure 1.2 CROSS SECTION OF RDK-408A2 COLD HEAD

2 MAINTENANCE

RDK-408A2 Cold Head is to be required to replace the sliding parts inside every 10,000 Hrs.

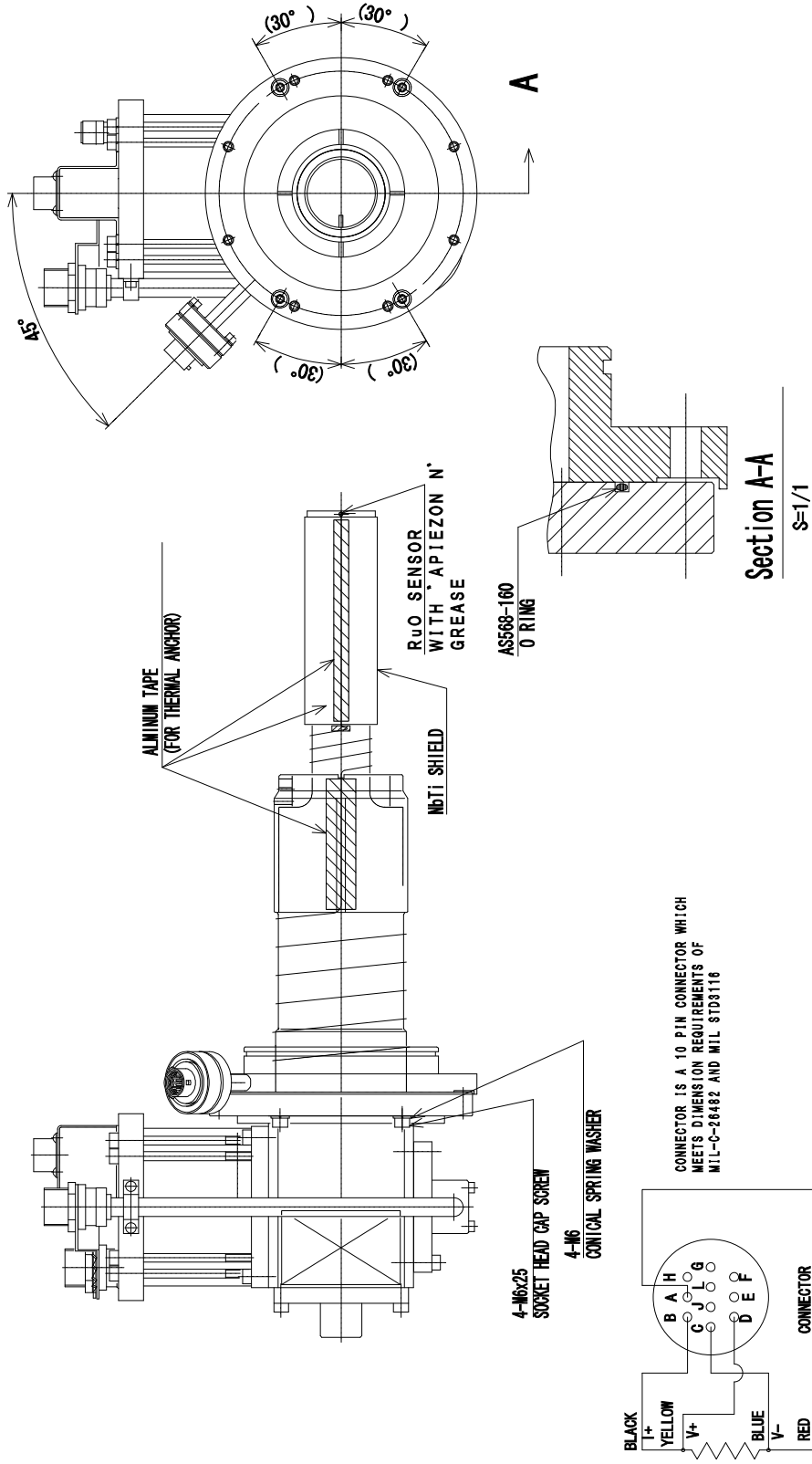
The maintenance work is not a Use's maintenance. Replace the Cold Head completely at site and return it to Sumitomo Heavy Industries, Ltd. for refurbishment.

APPENDIX**DRAWINGS**

No.	PART NAME	GE PART No.	SHI PART No.
1	RDK-408A2 COLD HEAD ASSEMBLY	2218465-3	RD42ZN1079
2	RDK-408A2 COLD HEAD	-	-

NOTE

1. ASSEMBLY IS TO BE CLEAN AND FREE FROM DIRT, OIL.
2. RUO SENSOR IS TO BE INSTALLED WITH 'APIEZON N' GREASE.
3. RUO SENSOR AND WIRING TO CONNECTOR PINS ARE TO BE INSPECTED.
4. OVERALL WIRE LENGTH APPROX. 1600mm.
5. ASSEMBLYS ARE TO BE BAGGED OR BOXED AND SEALED FROM DIRT AND MOISTURE.
6. ASSEMBLY IS TO BE MARKED WITH G. E. PART NUMBER, 2218465-3.



Section A-A

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RDK-408A2 COLD HEAD ASSEMBLY

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REVISION CONTROL

REVISION CONTROL

Manual No.	GE Revision	SHI Revision	Remarks	Date
CD32ZZ-098	Preliminary	-A	Publication of first edition.	MAY 16 / 2002
	Preliminary	-B	Change cooling capacity specification.	OCT. 31 / 2002
	Preliminary	-C	Change the outline drawing and SHI parts number.	JAN. 14 / 2003
	Revision 0	-D	This manual is included in the composite manual.	MAR. 17 / 2003



Sumitomo Heavy Industries, Ltd.

MANUAL NUMBER: CD32ZZ-058D

DATE: March 27 / 2003

TECHNICAL INSTRUCTION

RDK-408T 10K COLD HEAD Revision 2

For Service Personnel Only

PART NUMBERS

MODEL	SHI PART No.	GE PART No.
RDK-408T 10K COLD HEAD	RD42ZN0561	2200832

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FIGURE LIST

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1.2	CROSS SECTION OF THE RDK-408T COLD HEAD	6

CROSS REFERENCE

Before using this equipment, thoroughly read this manual and following manuals.

MANUAL NAME	MANUAL No.
OPERATION MANUAL SRDK Series CRYOCOOLER	CD32ZZ-056
TECHNICAL INSTRUCTION CSA-71A COMPRESSOR UNIT*	CD32ZZ-060
TECHNICAL INSTRUCTION CSW-71C COMPRESSOR UNIT*	CD32ZZ-061
TECHNICAL INSTRUCTION CSW-71D COMPRESSOR UNIT*	CD32ZZ-062
INSTALLATION MANUAL BPU-01 BY PASS UNIT**	CD32ZZ-075

* See the TECHNICAL INSTRUCTION of Compressor Unit used.

** See the INSTALLATION MANUAL of BPU-01, if the BPU-01 is used in this system.

1 GENERAL INFORMATION

The RDK-408T Cold Head is a two-stage GM cycle cryo-refrigerator. The function of the Cold Head is to produce continuous closed-cycle refrigeration at temperatures, depending upon the heat load imposed, in the range of 25 K to 40 K for the first-stage cold station and in the range of 6 K to 10 K for the second-stage cold station.

The Cold Head has three major components: the drive unit; the cylinder; and the displacer-regenerator assembly, which is located inside the cylinder.

With newly developed very unique 2nd stage structure, the model RDK-408T Cold Head has its 2nd stage refrigeration capacity of 5W at 10K.

Functionally, the high-pressure helium gas from the Compressor Unit will be supplied to the Cold Head through the helium gas supply connector. The supply gas will be passed into the displacer-regenerator assembly, come out through the displacer-regenerator assembly to the crankcase through the motor housing, and finally will be returned to the Compressor Unit through the helium gas return connector. The helium gas expansion in the displacer-regenerator assembly will be provided cooling condition for the first and second-stage cold stations.

1-1 SPECIFICATIONS

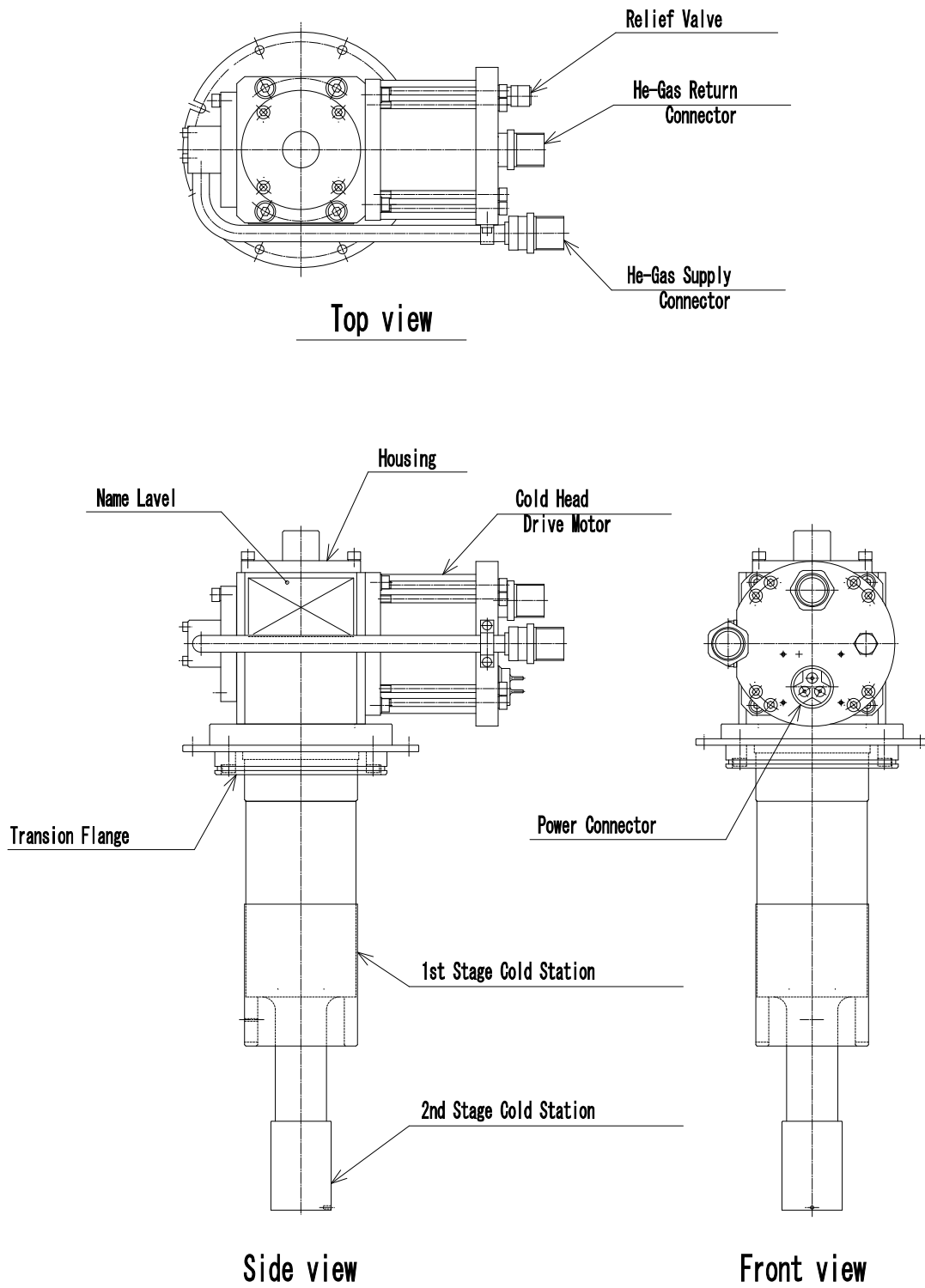
The specifications of the RDK-408T Cold Head are summarized in **Table 1.1**

Figure 1.1 shows the outline view of Cold Head.

Table 1.1 RDK-408T COLD HEAD SPECIFICATION

Refrigeration Capacity First Stage Second Stage	30 W at 45 K 5 W at 10 K
Orientation	Any Capacity Loss: Max. 15%
Ambient Operating Temperature	5 to 28 deg.C (41 to 82.4 deg.F) 28 to 35 deg.C (82.4 to 95 deg.F) with 5% Capacity Loss
Helium Gas Pressure Static Operating (High Side)*	1.45 - 1.50 MPa at 20 deg.C (68 deg.F) (14.8 - 15.3 kgf/cm ² G, 210 - 217 psig) 2.10 - 2.30 MPa --- approx. (21.4 - 23.5 kgf/cm ² G, 319 - 333 psig)
Pressure Relief Valve Setting	1.86 - 1.96 MPa (19.0 - 20.0 kgf/cm ² G, 270 - 284 psig)
Gas Supply Connector Gas Return Connector	1/2-inch Coupling 1/2-inch Coupling
Dimension Width Length Height	180 mm (7.09') 288 mm (11.33') 520 mm (20.47')
Weight	17.2 kg (37.9 LBS) --- approx.

* The operating pressure varies according to the heat load of cold head and temperature around the equipment.



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Figure 1.1 OUTLINE VIEW OF COLD HEAD MODEL RDK-408T

1-2 CONSTRUCTION

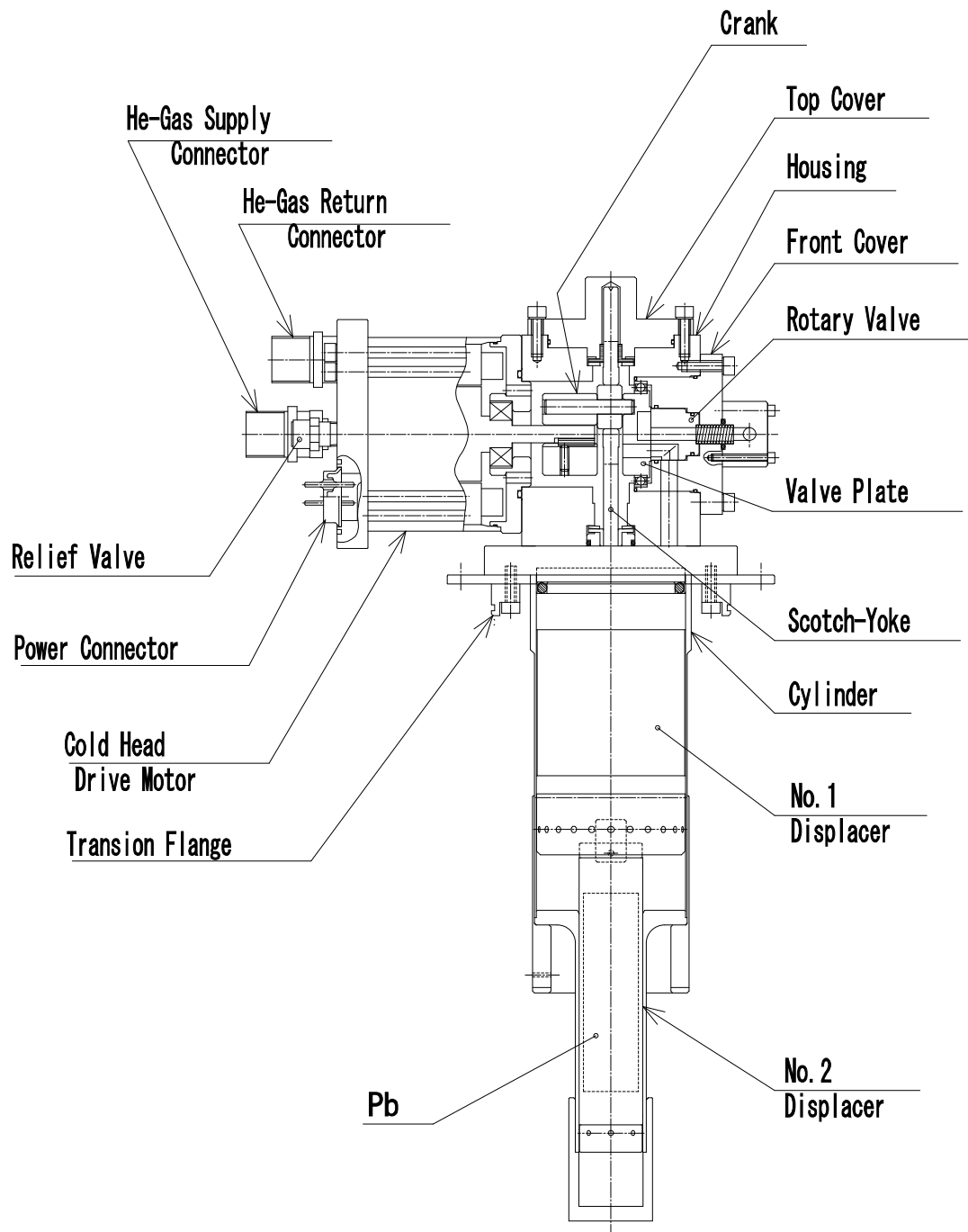
The cross section of the RDK-408T Cold Head is shown in **Figure 1.2**. It consists of a Cylinder, No.1 Displacer, No.2 Displacer, drive mechanism, and Cold Head Drive Motor. No.1 Displacer is connected to the Scotch Yoke which can be driven by the Cold Head Drive Motor through the Crank with Bush so that the rotation of the Cold Head Drive Motor can be varied to reciprocating motion of Scotch Yoke and Displacers.

The Rotary Valve system is furnished to control the helium gas intake and exhaust timing. The Rotary Valve is also coupled to the Cold Head Drive Motor through Crank, so intake and exhaust operation is synchronized with the position of the Displacer.

The Displacer is a loose fit in the Cylinder except at the top and where it is equipped with a dynamic(sliding) seal to prevent leakage passed through the clearance between the Displacer and Cylinder.

The Displacers consist of regenerator material which cool the gas when passing downwards to the cold space and heats the gas when passing upwards from the cold space.

The pressure above and below the Displacer is the same except for small pressure drops across the regenerator when gas is flowing through it. Virtually no physical work is required to move the Displacer in the Cylinder. No work is done on the gas and the gas does no work on the Displacer. The pressure in the system is increased or decreased by operation of the inlet or outlet valves.



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Figure 1.2 CROSS SECTION OF THE RDK-408T COLD HEAD

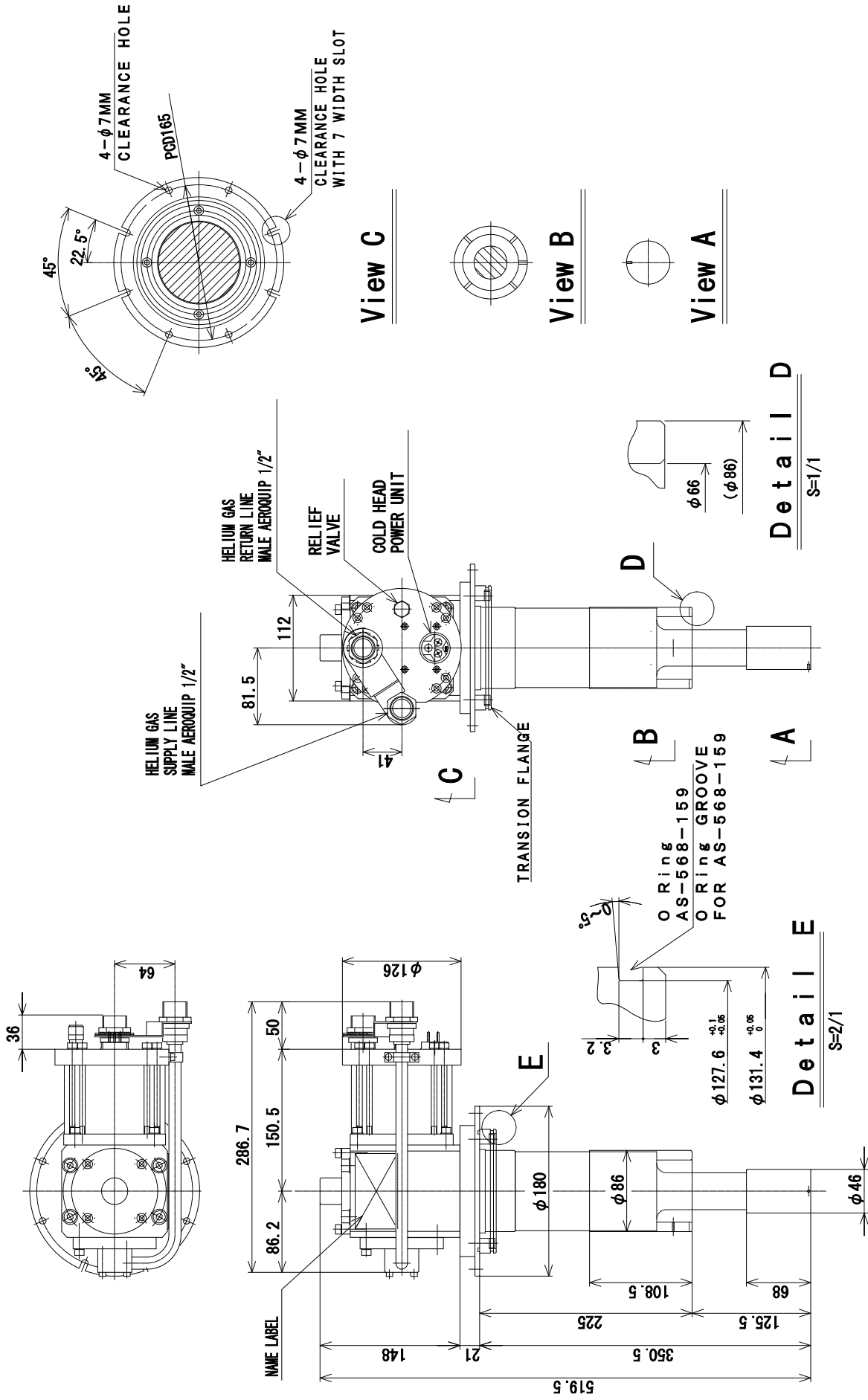
2 MAINTENANCE

The RDK-408T Cold Head is to be required to replace the sliding parts inside every 10,000 Hrs.

The maintenance work is not a Use's maintenance. Replace the Cold Head completely at site and return it to Sumitomo Heavy Industries, Ltd. for refurbishment.

APPENDIX**DRAWINGS**

No.	PART NAME	GE PART No.	SHI PART No.
1	RDK-408T COLD HEAD	2200832	RD42ZN0561



RDK-408T COLD HEAD

XRD42Y0178A0

REVISION CONTROL

REVISION CONTROL

Manual No.	GE Revision	SHI Revision	Remarks	Date
CD32ZZ-058	Revision 2	-A	Publication of first edition.	DEC. 20 / 1999
	Revision 2	-B	Add the description of BPU-01.	MAR. 30 / 2000
	Revision 2	-C	Change the SHI address.	JAN. 25 / 2001
	Revision 2	-D	Correct the Spec. of 1 st Stage Cooling Capacity.	MAR. 27 / 2003